OSSD - Calculus and Vector

MCV4U

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Unit 1: Rate of change. In the first unit, we will discuss the concept of Rate of Change, Continuity and Limits. Also, the the introduction to the Derivatives will be discussed.

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1 Slope

2 Derivatives

2.1 Introduction to Derivatives

Definition 2.1 (Derivative)**.** is a fundamental tool that measures the instantaneous rate of change of a function with respect to its input, representing the slope of the tangent line to a graph at a given point.
You can write Derivative in three ways:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$m_T = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

The simple Notation:

- Average rate of change in y values over an interval of x values: $\frac{\Delta y}{\Delta x}$
- Instantaneous rate of change in the y value at an x value: $\frac{\delta y}{\delta x}$ or $\frac{dy}{dx}$.

When you solve for the instantaneous rate of change for function $f(x)$ at $x = a$. You can write the Derivative in one of two ways.

$$\left.\frac{dy}{dx}\right|_{x=a} = \frac{f(a+h) - f(a)}{h}$$

$$f'(a) = \frac{f(a+h) - f(a)}{h}$$

There is also a different way to solve for m_T
Given we have a function $f(x)$, we grad two points $(a, f(a))$ and $(x, f(x))$ The instantaneous rate of change of the function should be

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

Example 2.2. Given the function $f(x) = \frac{1}{x}$, find the slope of the tangent at $x = 2$
Take the point $(2, \frac{1}{2})$ and $(x, \frac{1}{x})$

$$\begin{aligned}
m &= \lim_{x \to 2} \frac{\Delta x}{\Delta y} && (1)\\
&= \lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} && (2)\\
&= \cdots && (3)\\
&= \lim_{x \to 2} \frac{-x+2}{2x} * \frac{1}{x-2} && (4)\\
&= \lim_{x \to 2} -\frac{1}{2x} && (5)\\
&= \frac{1}{4} && (6)
\end{aligned}$$

The derivative of a function cannot exist if:

- There is a discontinuity at the given x value
- The function is continuous at a given "x" value but
  - You have a "corner" at the "x" value
  - You have a "cus" at the "x" value
  - You have a vertical tangent at the "x' value

2.2 The Derivatives of Polynomial Functions

2.2.1 What is a power?

To start off, we need to define the power function.

$$x^a, x \in \mathbb{R}, a \in \mathbb{R}$$

At here, x is the base, and a is the exponent. Base and exponent together is called "Power".

$$x^a$$

should be read as:

"x to a"

2.2.2 Power Rules

If

$$f(x) = x^n$$

the derivative of $f(x)$ is

$$f'(x) = nx^{n-1}$$

In words:

"To find the derivative of a power function, take the exponent from the power function and use it as the coefficient of the derivative. Then reduce the exponent value for the derivative by 1 from the original function."

Remark. Do not use "Power Rule" for exponent function (a^x), such as 2^x

2.2.3 Derivative Rules

$$\text{If } f(x) = c,\ f'(x) = 0$$
$$\text{If } f(x) = x^n, f'(x) = nx^{n-1}$$
$$\text{If } f(x) = cg(x), f'(x) = c * g'(x)$$
$$\text{If } h(x) = f(x) + g(x), h'(x) = f'(x) + g'(x)$$
$$\text{If } h(x) = f(x) - g(x), h'(x) = f'(x) - g'(x)$$

3 Reference Sheet

3.1 Assume Knowledge

3.1.1 Log Laws

$$\log_b A + \log_b B = \log_b(A \times B)$$
$$\log_b(\tfrac{M}{N}) = \log_b M - \log_b N$$
$$log_b M^k = k \log_b M$$
$$log_b b^k = k$$
$$b^{\log_b k} = k$$

3.1.2 Trig Laws

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$
$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$
$$\csc\theta = \frac{1}{\sin\theta}$$
$$\sec\theta = \frac{1}{\cos\theta}$$
$$\sin^2\theta + \cos^2\theta = 1$$
$$1 + \cot^2\theta = \csc^2\theta$$
$$\tan^2\theta + 1 = \sec^2\theta$$
$$\sin(A \pm B) = \sin A\cos B \pm \sin B\cos A$$
$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

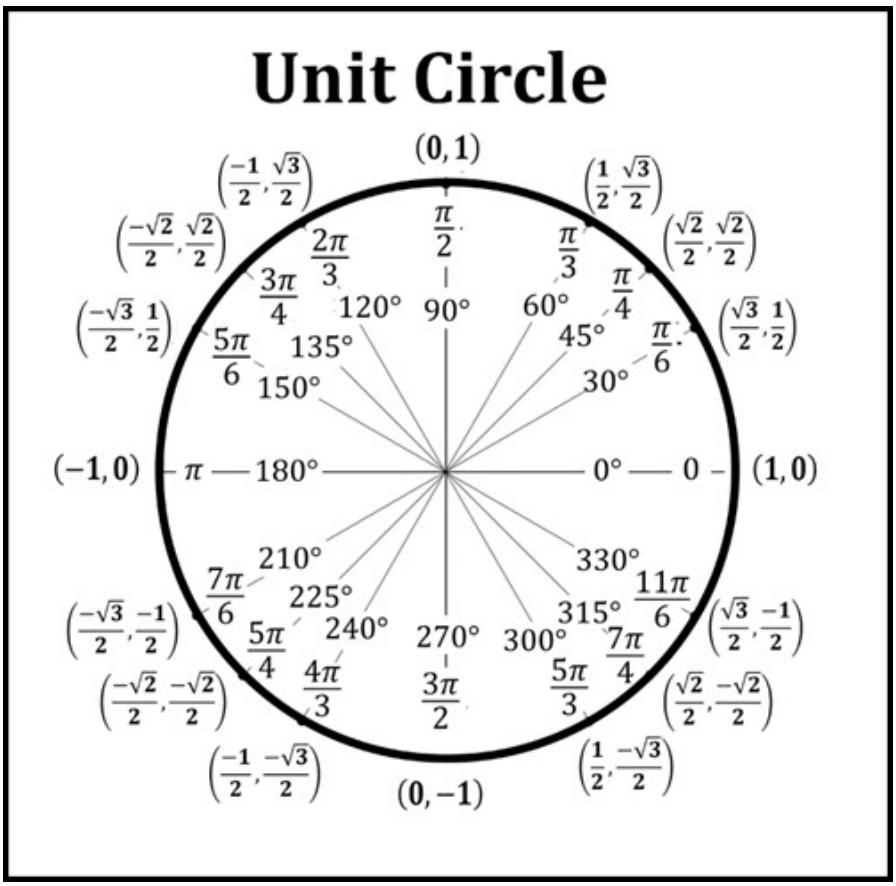
sin Law:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Cosine Law:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Unit Circle:


Unit Circle
(0, 1)
(1, 0)
(−1, 0)
(0, −1)
π/2
90°
π/3
60°
π/4
45°
π/6
30°
0°
0
2π/3
120°
3π/4
135°
5π/6
150°
π
180°
7π/6
210°
5π/4
225°
4π/3
240°
3π/2
270°
5π/3
300°
7π/4
315°
11π/6
330°
(1/2, √3/2)
(√2/2, √2/2)
(√3/2, 1/2)
(−1/2, √3/2)
(−√2/2, √2/2)
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